# PIR - Performance Indicator Record

Review identification, purpose and boundary

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| | | | |
|---|---|---|---|
| **Study / protocol**<br>GLENOS-01 / GS-2026-01 | **CIP version**<br>CIP v4.2 | **Site / code**<br>Riverside Orthopedic Center / 014 | **PIR ID**<br>PIR-014-2026-10 |
| **Period reviewed / data cut-off**<br>1 to 31 Oct 2026 / 31 Oct 2026 | **Previous PIR**<br>PIR-014-2026-09 | **Rule set (version, effective date)**<br>PIR-RS-014 v1, 12 Aug 2026 | **Review date**<br>5 Nov 2026 |
| **Review type**<br>☒ Routine ☐ Early cases<br>☐ Triggered ☐ Action check | **Prepared by / role**<br>Maria Silva / Study Coordinator | **Accountable reviewer / role**<br>Dr Alex Morgan / PI | **Controlled location**<br>ISF tab 3 (Quality) / PIR-014-2026-10 |

## A Purpose and use

The PIR reviews the site's performance over one period through six indicators fixed before the first participant, plus any the site adds. It takes the results already reported by the source records (FUCR, SLCT, CSST, IMC, SART and others) and does not recount them. What it adds is the reading across time and across indicators: how each result moves over the last periods (D), and whether problems in different indicators share the same participants, surgeon, staff, time or process step (E). From that it decides what to correct (F), checks whether earlier actions worked (G) and concludes on the period (H).

Every result keeps its count and denominator; a percentage alone hides small numbers. A rule is never changed to fit a result; a new rule applies from its effective date.

| Unit of analysis | Decision supported | Inputs | Outputs |
|---|---|---|---|
| One site, one review period, each indicator with its own denominator. | Whether each indicator is within its rule, needs closer review or action, and what the site must improve. | Results of the source records; the previous PIR; the rule set. | Trend and links (D, E); improvement actions with a check (F, G); routes to CAPA, SRR, POT, SLCT or the sponsor. |

## B Review boundary

Fix the boundary before reading any result.

| | |
|---|---|
| **Records included (IDs or ranges, cut-off)** | SIFSL-014 rev. 23; PSEAR-014-019 to -022; ICDL-014-007, -008; FUCR-014-2026-10; SLCT S-01 to S-03; CSST-014-02; IMC-014-02; SART-014-004, -005; query report 31 Oct |
| **Records missing, late or excluded, and the effect on denominators** | FUCR for October closed 3 Nov (late); its counts are final. No record excluded. |
| **Rule or source changes in the period** | None. Rule set v1 in force for July to October. |
| **Periods compared and whether they are comparable** | July to October, same rules and units; July and August have few cases |

# PIR - Performance Indicator Record

Indicator rules (rule sheet)

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| Rule set ID / version | Effective from | Approved by (PI) / date | Replaces version |
|---|---|---|---|
| PIR-RS-014 / v1 | 12 Aug 2026 | Dr A. Morgan / 11 Aug 2026 | None (first version) |

## C Indicator rules (rule sheet)

Filled once before the first participant and signed by the PI; the same sheet is attached to every PIR. A rule changes only by a new version entered in the change log, never because of a result already seen. Rows 7 and 8: indicators the site adds.

| Indicator | Source records | Unit (denominator) | Within rule when | Closer review when | Action required when | Rule ID |
|---|---|---|---|---|---|---|
| **1 · Enrollment quality** | PSEAR, SIFSL, deviation record | Inclusion decisions | Every criterion evidenced before the decision; no important eligibility deviation | A clarification after enrollment | Any important eligibility deviation | PIR-01 |
| **2 · Protocol deviations** | Deviation record, ICDL, IMC | Procedures | No important deviation of the same type in two periods | One important deviation in the period | Same type again after a correction | PIR-02 |
| **3 · Follow-up compliance** | FUCR, PFVR | Expected visits | In window with all components: 90% or more | 80 to 89%, or worse two periods in a row | Below 80% | PIR-03 |
| **4 · Safety reporting timeliness** | SART | Reportable cases | All sent within the CIP interval | One complete first report missing | Any late report | PIR-04 |
| **5 · Surgeon procedural consistency** | SLCT, ICDL, CSST | Procedures per surgeon | At most one unplanned event (substitution, deviation, unplanned support, grade IIIa or higher) in a surgeon's last 5 cases | Two unplanned events in a surgeon's last 5 cases | Three or more in the last 5 cases, or a grade IIIa or higher complication repeated | PIR-05 |
| **6 · Critical data completeness** | IMC, query log, operative notes | Endpoint fields sampled | Agreement 97% or more; fewer than 5 queries open over 10 days | Agreement 95 to 96.9%, or 5 or more queries over 10 days | Agreement below 95% | PIR-06 |
| | | | | | | |
| | | | | | | |

**Change log**

| Version | Effective from | What changed and why | Approved by (PI) / date |
|---|---|---|---|
| | | | |
| | | | |

# PIR - Performance Indicator Record

Results across periods

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| PIR ID | Site / code | Period / data cut-off | Rule set (version) |
|---|---|---|---|
| PIR-014-2026-10 | Riverside / 014 | Oct 2026 / 31 Oct | PIR-RS-014 v1 |

## D Results across periods

Copy each result from its source record as count / denominator (%). Direction compares the last four periods: a change counts when it passes the closer-review margin or lasts two periods in a row; with fewer than three periods write Too few periods.

| Indicator | Three periods back | Two periods back | Last period | This period | Cumulative | Direction | Decision (rule in C) |
|---|---|---|---|---|---|---|---|
| **1 · Enrollment quality** | 3 / 3 (100%) | 2 / 2 (100%) | 1 / 1 (100%) | 0 / 0 | 8 / 8 (100%) | ☐ Better<br>☒ Stable<br>☐ Worse<br>☐ Too few periods | ☒ Within rule<br>☐ Closer review<br>☐ Action required<br>☐ Not assessable |
| **2 · Protocol deviations** | 1 / 3 | 0 / 1 | 0 / 2 | 2 / 2 | 3 / 8 | ☐ Better<br>☐ Stable<br>☒ Worse<br>☐ Too few periods | ☐ Within rule<br>☐ Closer review<br>☒ Action required<br>☐ Not assessable |
| **3 · Follow-up compliance** | 3 / 3 (100%) | 5 / 6 (83%) | 6 / 7 (86%) | 5 / 7 (71%) | 19 / 23 (83%) | ☐ Better<br>☐ Stable<br>☒ Worse<br>☐ Too few periods | ☐ Within rule<br>☐ Closer review<br>☒ Action required<br>☐ Not assessable |
| **4 · Safety reporting timeliness** | 0 / 0 | 0 / 0 | 1 / 1 | 1 / 1 | 2 / 2 | ☐ Better<br>☐ Stable<br>☐ Worse<br>☒ Too few periods | ☐ Within rule<br>☐ Closer review<br>☐ Action required<br>☒ Not assessable |
| **5 · Surgeon procedural consistency** | 1 / 3 | 0 / 1 | 1 / 2 | 1 / 2 | S-01 0 / 3; S-02 1 / 3; S-03 2 / 2 | ☐ Better<br>☒ Stable<br>☐ Worse<br>☐ Too few periods | ☐ Within rule<br>☒ Closer review<br>☐ Action required<br>☐ Not assessable |
| **6 · Critical data completeness** | 60 / 60 | 118 / 118 | 120 / 120 | 116 / 118 (98.3%) | 414 / 416 (99.5%) | ☐ Better<br>☐ Stable<br>☒ Worse<br>☐ Too few periods | ☒ Within rule<br>☐ Closer review<br>☐ Action required<br>☐ Not assessable |
| | | | | | | ☐ Better<br>☐ Stable<br>☐ Worse<br>☐ Too few periods | ☐ Within rule<br>☐ Closer review<br>☐ Action required<br>☐ Not assessable |
| | | | | | | ☐ Better<br>☐ Stable<br>☐ Worse<br>☐ Too few periods | ☐ Within rule<br>☐ Closer review<br>☐ Action required<br>☐ Not assessable |

# PIR - Performance Indicator Record

Links between indicators

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| PIR ID | Site / code | Period / data cut-off | Rule set (version) |
|---|---|---|---|
| PIR-014-2026-10 | Riverside / 014 | Oct 2026 / 31 Oct | PIR-RS-014 v1 |

## E Links between indicators

For every indicator not within its rule, look for the same participants, surgeon, staff member, time point, component or process step in the other indicators and in the source records. Write what was compared and the counts. A shared pattern is a lead to test, not a proof of cause.

| # | Pattern seen | Indicators involved | Source records (IDs and sections) | Comparison and counts | Likely shared cause | Conclusion |
|---|---|---|---|---|---|---|
| 1 | Instability events concentrate in cases with a missing trial liner or an unplanned liner change | 2, 5 and CSST | ICDL-014-002 F-1; ICDL-014-005 F; ICDL-014-007 F; CSST-014-02 C2 and E2; SLCT S-02, S-03 | Such cases: 3 (002, 005, 007), instability in 2. Other cases: 5, instability in 0 | Loaner-set check sheet does not list trial liners; not the surgeon alone | ☒ Supported<br>☐ Not supported<br>☐ Open: needs more data |
| 2 | Follow-up gaps and transcription errors in the same weeks | 3 and 6 | FUCR-014-2026-10; PFVR-014-002-M3 H; IMC-014-02 checks 8 and 9 | Issues on visits 1 to 14 Oct (nurse on leave): 3 of 4. Rest of the period: 1 of 4 | No cover plan for questionnaires and data entry during absence | ☒ Supported<br>☐ Not supported<br>☐ Open: needs more data |
| 3 | Both instability events in fracture-sequelae cases | 1, 5 and CSST | CSST-014-02 E3; PSEAR of both participants (indication) | Fracture sequelae: 2 cases, instability in 2. Other types: 6 cases, instability in 0 | Case type; numbers too small | ☐ Supported<br>☐ Not supported<br>☒ Open: needs more data |
| 4 | | | | | | ☐ Supported<br>☐ Not supported<br>☐ Open: needs more data |

# PIR - Performance Indicator Record

Improvement actions and their effect

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|---|---|---|---|
| PIR-014-2026-10 | Riverside / 014 | Oct 2026 / 31 Oct | PIR-RS-014 v1 |

## F Improvement actions

One action per cause found in D or E. Set the check that will show it worked before starting it. The receiving record owns the action.

| Action | What will change (from D or E) | Receiving record | Owner | Due | How we will know it worked (indicator, value, period) |
|---|---|---|---|---|---|
| PIR-A1 | Trial liners on the loaner-set check sheet; no case starts without them (E-1) | CAPA-014-04; SRR R-09 | OR nurse lead | 15 Nov | Indicator 2: no trial-less case in the next 5 procedures |
| PIR-A2 | Cover plan for questionnaires and data entry when the research nurse is away (E-2) | SRR R-13; delegation log | Coordinator | 30 Nov | Indicator 3 at 90% or more and indicator 6 at 99% or more in Nov and Dec |
| PIR-A3 | Report instability by case type to the sponsor at the next safety call (E-3) | Sponsor call minutes | PI | 20 Nov | Sponsor answer recorded; case type added to the next CSST |
| | | | | | |

## G Did earlier actions work?

Compare the indicator before and after with the check set in F. Done is not the same as effective.

| Action | Check set when assigned | Value before | Value now | Result | Next step |
|---|---|---|---|---|---|
| ICDL J-1 | Sponsor completes the loaner sets (July) | Jul: 1 trial-less case | Oct: 1 trial-less case | ☐ Worked<br>☒ Did not work<br>☐ Too early | Replaced by PIR-A1 (site check, not sponsor request) |
| IMC A-01-03 | No late image set in two months | Jul: 2 late | Sep, Oct: 0 late | ☒ Worked<br>☐ Did not work<br>☐ Too early | Close |
| | | | | ☐ Worked<br>☐ Did not work<br>☐ Too early | |
| | | | | ☐ Worked<br>☐ Did not work<br>☐ Too early | |

# PIR - Performance Indicator Record

Period conclusion

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|---|---|---|---|
| PIR-014-2026-10 | Riverside / 014 | Oct 2026 / 31 Oct | PIR-RS-014 v1 |

## H Period conclusion

The conclusion covers only the stated period, records and indicators; it does not certify the site.

| Decision | Count | Indicators (number and name) |
|---|---|---|
| **Within rule** | 2 | 1 · Enrollment quality; 6 · Critical data completeness (worse, still within rule) |
| **Closer review** | 1 | 5 · Surgeon procedural consistency (S-03: 2 unplanned events in 2 cases) |
| **Action required** | 2 | 2 · Protocol deviations; 3 · Follow-up compliance |
| **Not assessable** | 1 | 4 · Safety reporting timeliness (only 2 cases so far) |

| | |
|---|---|
| **Site performance over the periods** | ☐ Improving ☐ Stable ☐ Declining ☒ Mixed |
| **What the site learned this period** | Two indicators need action and one needs closer review in October, and they trace back to two process causes: loaner sets without trial liners (indicators 2 and 5, and the instability in the CSST) and no cover when the research nurse is away (indicator 3, and the small drop in indicator 6). The July request to the sponsor did not fix the sets; the fix has to be the site's own check before each case. |
| **Routes opened (CAPA, SRR, POT, SLCT, sponsor) or none** | CAPA-014-04; SRR R-09 and R-13 updated; POT-014-08 (S-03 proctoring continues); sponsor call 20 Nov |
| **Next review (date or trigger)** | PIR-014-2026-11, 5 Dec 2026; earlier if a trial-less case occurs |

**Prepared by**

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Signature / date:

*M. Silva* 5 Nov 2026

**Principal Investigator**

Name: Dr Alex Morgan

Signature / date:

*A. Morgan* 6 Nov 2026